15	**D**	At time t = 0, object is at the extreme end of its SHM. At time t = T/4, object is at equilibrium position. At time t = T/2, object is at the other extreme end. Since speed is maximum at equilibrium position, mass has maximum speed at time t = T/4.
16	**B**	Total energy $E = \frac{1}{2}m\omega^2 A^2$